

Containerization in DevOps

Lab – 1

Linux Process Commands

Name	Arnav Rajput
Enrollment No.	S24BCAU0194
Lab	Containerization in DevOps – Lab 1
Platform	WSL (Ubuntu/Linux environment)

1. Objective

To practice Linux process-management commands in a WSL environment, including process listing, background and foreground jobs, terminating processes, process-tree inspection, system monitoring, and process priority.

2. Commands Performed

sleep 500 — Starts a sleep process for 500 seconds.

Ctrl + Z — Suspends the foreground process and returns control to the shell.

ps -aux — Displays detailed information about running processes.

gedit test.txt & — Opens a text file in Gedit and runs it in the background.

kill — Terminates the process identified by its PID.

ps -p — Checks the status of a specific process.

bg — Resumes a stopped job in the background.

fg — Attempts to bring a background job to the foreground.

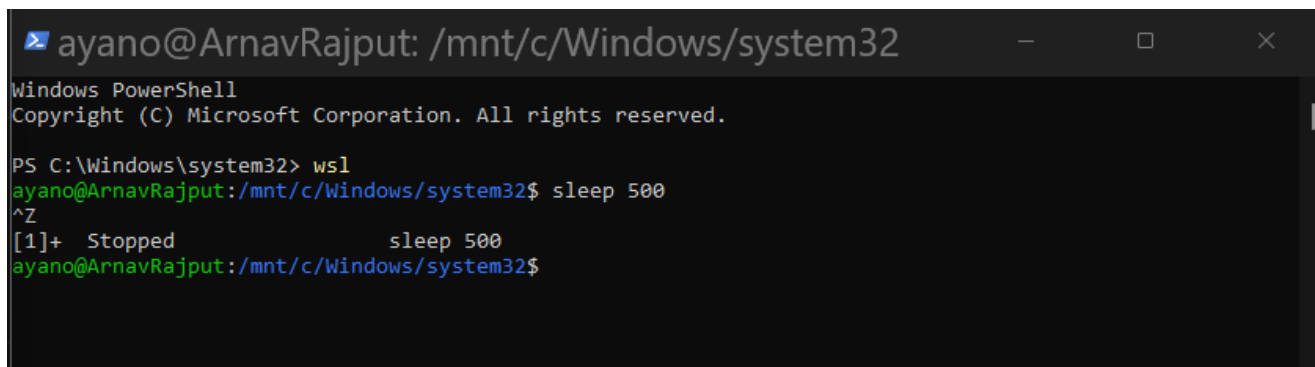
pstree — Displays processes in a tree/hierarchical structure.

top — Displays live process, CPU, and memory information.

nice -n 10 sleep 60 & — Starts a process with a nice value of 10.

3. Practical Output Screenshots

1. sleep 500 and Ctrl+Z



```
ayano@ArnavRajput: /mnt/c/Windows/system32
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

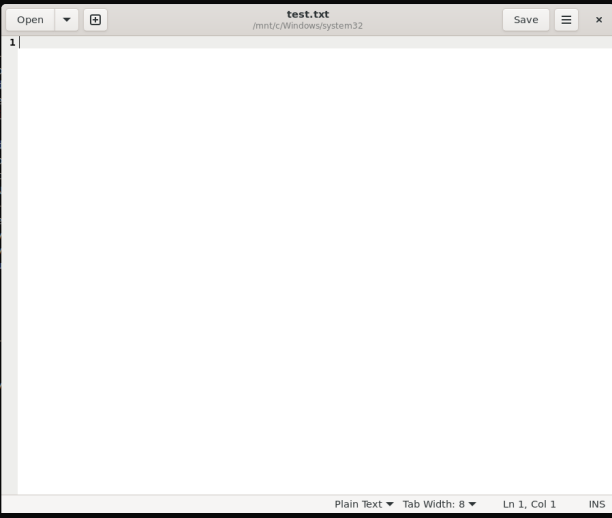
PS C:\Windows\system32> wsl
ayano@ArnavRajput:/mnt/c/Windows/system32$ sleep 500
^Z
[1]+  Stopped                  sleep 500
ayano@ArnavRajput:/mnt/c/Windows/system32$
```

2. ps -aux

```
ayano@ArnavRajput:/mnt/c/windows/system32$ ps -aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.7  0.3 21804 13156 ?        Ss   09:03   0:01 /sbin/init
root         2  0.0  0.0  3180  2204 hvcc0   Sl+  09:03   0:00 /init
root         7  0.0  0.0  3180  2044 hvcc0   Sl+  09:03   0:00 plan9 --control-socket 7 --log-leve
root        43  0.2  0.4 50444 15900 ?        S<s  09:03   0:00 /usr/lib/systemd/systemd-journald
root        90  0.1  0.1 25500  6928 ?        Ss   09:03   0:00 /usr/lib/systemd/systemd-udev
systemd+   109  0.0  0.3 21468 13360 ?        Ss   09:03   0:00 /usr/lib/systemd/systemd-resolved
systemd+   117  0.0  0.2  91036  8056 ?        Ssl  09:03   0:00 /usr/lib/systemd/systemd-timesyncd
root       158  0.0  0.0  4244  2744 ?        Ss   09:03   0:00 /usr/sbin/cron -f -P
message+   159  0.0  0.1  9636  5396 ?        Ss   09:03   0:00 @dbus-daemon --system --address=sys
root       166  0.0  0.2 17980  8952 ?        Ss   09:03   0:00 /usr/lib/systemd/systemd-logind
root       169  0.0  0.3 1756628 13736 ?       Ssl  09:03   0:00 /usr/libexec/wsl-pro-service
syslog     177  0.0  0.1 222516  6084 ?        Ssl  09:03   0:00 /usr/sbin/rsyslogd -n -iNONE
root       182  0.0  0.0  3124  2036 tty1     Ss+  09:03   0:00 /sbin/agetty -o -p -- \u --noclear
root       199  0.1  0.6 112744 25148 ?        Ssl  09:03   0:00 /usr/bin/python3 /usr/share/unatten
root       291  0.0  0.0  2808  2016 ?        Ss   09:03   0:00 /bin/sh /usr/lib/apt/apt.systemd.da
root       306  0.0  0.0  2808  2028 ?        S    09:03   0:00 /bin/sh /usr/lib/apt/apt.systemd.da
root       380 43.1  3.1 363544 123004 ?       Sl   09:03   0:57 /usr/bin/python3 /usr/bin/unattende
root       531  0.0  0.0  3188  1108 ?        Ss   09:03   0:00 /init
root       532  0.0  0.0  3204  1248 ?        S    09:03   0:00 /init
ayano      533  0.0  0.1  6080  5372 pts/0    Ss   09:03   0:00 -bash
root       534  0.0  0.1  6704  4740 pts/1    Ss   09:03   0:00 /bin/login -f
ayano      580  0.1  0.3 20332 11644 ?        Ss   09:04   0:00 /usr/lib/systemd/systemd --user
ayano      581  0.0  0.0 21160  3572 ?        S    09:04   0:00 (sd-pam)
ayano      607  0.0  0.1  6080  5328 pts/1    S+   09:04   0:00 -bash
root       715  0.0  0.0  2808  2024 ?        Ss   09:04   0:00 /bin/sh /usr/lib/apt/apt.systemd.da
root       718  0.0  0.0  3124  1900 ?        S    09:04   0:00 flock -w 3600 3
ayano      738  0.0  0.0  3132  1964 pts/0    T    09:04   0:00 sleep 500
_apt       741  0.2  0.2 20324 10768 ?        S    09:04   0:00 /usr/lib/apt/methods/http
_apt       742  0.3  0.2 20324 10884 ?        S    09:04   0:00 /usr/lib/apt/methods/http
ayano      762  0.0  0.1  9588  5160 pts/0    R+   09:06   0:00 ps -aux
ayano@ArnavRajput:/mnt/c/windows/system32$
```

3. gedit running in background

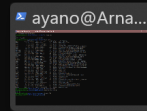
```
ayano@ArnavRajput: /mnt/c/Windows/system32
PS C:\Windows\system32> wsl
ayano@ArnavRajput:/mnt/c/Windows/system32$ sleep 500
^Z
[1]+  Stopped                  sleep 500
ayano@ArnavRajput:/mnt/c/Windows/system32$ ps -aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.7  0.3 21804 13156 ?        Ss   09:03   0:01 /sbin/init
root         2  0.0  0.0   3180  2204 hvc0     Sl+  09:03   0:00 /init
root         7  0.0  0.0   3180  2044 hvc0     Sl+  09:03   0:00 plan9 --control-socket 7 --
root        43  0.2  0.4 50444 15900 ?        S<s  09:03   0:00 /usr/lib/systemd/systemd-jou
root        90  0.1  0.1 25500  6928 ?        Ss   09:03   0:00 /usr/lib/systemd/systemd-ucr
systemd+  109  0.0  0.3 21468 13360 ?        Ss   09:03   0:00 /usr/lib/systemd/systemd-rs
systemd+  117  0.0  0.2 91036  8056 ?        Ssl  09:03   0:00 /usr/lib/systemd/systemd-ti
root       158  0.0  0.0   4244  2744 ?        Ss   09:03   0:00 /usr/sbin/cron -f -p
message+  159  0.0  0.1   9636  5396 ?        Ss   09:03   0:00 @dbus-daemon --system --add
root      166  0.0  0.2 17980  8952 ?        Ss   09:03   0:00 /usr/lib/systemd/systemd-ld
root      169  0.0  0.3 1756628 13736 ?      Ssl  09:03   0:00 /usr/libexec/wsl-pro-servic
syslog    177  0.0  0.1 222516  6084 ?        Ssl  09:03   0:00 /usr/sbin/rsyslogd -n -iNO
root      182  0.0  0.0   3124  2036 tty1     Ss+  09:03   0:00 /sbin/agetty -o -p -- \u --
root      199  0.1  0.6 112744 25148 ?        Ssl  09:03   0:00 /usr/bin/python3 /usr/share
root      291  0.0  0.0   2808  2016 ?        Ss   09:03   0:00 /bin/sh /usr/lib/apt/apt.sy
root      306  0.0  0.0   2808  2028 ?        S   09:03   0:00 /bin/sh /usr/lib/apt/apt.sy
root     380 43.1  3.1 363544 123004 ?      Sl   09:03   0:57 /usr/bin/python3 /usr/bin/u
root      531  0.0  0.0   3188  1108 ?        Ss   09:03   0:00 /init
root      532  0.0  0.0   3204  1248 ?        S   09:03   0:00 /init
ayano     533  0.0  0.1   6080  5372 pts/0    Ss   09:03   0:00 -bash
root      534  0.0  0.1   6704  4740 pts/1    Ss   09:03   0:00 /bin/login -f
ayano     580  0.1  0.3 20332 11644 ?        Ss   09:04   0:00 /usr/lib/systemd/systemd --
ayano     581  0.0  0.0   2160  3572 ?        S   09:04   0:00 (sd-pam)
ayano     607  0.0  0.1   6080  5328 pts/1    S+   09:04   0:00 -bash
root      715  0.0  0.0   2808  2024 ?        Ss   09:04   0:00 /bin/sh /usr/lib/apt/apt.sy
root      718  0.0  0.0   3124  1900 ?        S   09:04   0:00 flock -w 3600 3
ayano     738  0.0  0.0   3132  1964 pts/0    T   09:04   0:00 sleep 500
lapt      741  0.2  0.2 20324 10768 ?        S   09:04   0:00 /usr/lib/apt/methods/http
lapt      742  0.3  0.2 20324 10884 ?        S   09:04   0:00 /usr/lib/apt/methods/http
ayano     762  0.0  0.1   9588  5160 pts/0    R+   09:06   0:00 ps -aux
ayano@ArnavRajput:/mnt/c/windows/system32$ touch test.txt
ayano@ArnavRajput:/mnt/c/windows/system32$ gedit test.txt &
[2] 796
ayano@ArnavRajput:/mnt/c/windows/system32$
```



The screenshot shows a Windows terminal window with a WSL session. The terminal output displays the results of a `ps -aux` command, listing various system processes and user processes. A `sleep 500` command was interrupted with `^Z`, and the `ps -aux` command was run. The output shows processes like `/sbin/init`, `/usr/lib/systemd/systemd-journald`, `/usr/lib/systemd/systemd-udevd`, `/usr/lib/systemd/systemd-rs`, `/usr/lib/systemd/systemd-timedatectl`, `/usr/sbin/cron`, `@dbus-daemon`, `/usr/lib/systemd/systemd-ld`, `/usr/libexec/wsl-pro-service`, `/usr/sbin/rsyslogd`, `/sbin/agetty`, `/usr/bin/python3`, `/bin/sh`, `/usr/lib/apt/apt.systemd.daily`, `/usr/bin/python3`, `/init`, `/bin/login`, `/usr/lib/systemd/systemd`, `(sd-pam)`, `/bin/sh`, `/usr/lib/apt/apt.systemd.daily`, `flock`, `sleep`, `/usr/lib/apt/methods/http`, and `ps -aux`. The terminal also shows the execution of `touch test.txt` and `gedit test.txt &`. In the background, a `test.txt` window is visible, showing the file's content.

4. kill Gedit process

```
ayano@ArnavRajput:/mnt/c/windows/system32$ touch test.txt  
ayano@ArnavRajput:/mnt/c/windows/system32$ gedit test.txt &  
[2] 796  
ayano@ArnavRajput:/mnt/c/windows/system32$ kill 796  
ayano@ArnavRajput:/mnt/c/windows/system32$
```



5. Verify Gedit termination

```
ayano@ArnavRajput:/mnt/c/Windows/system32$ ps -p 796
  PID TTY          TIME CMD
[2]-  Terminated                  gedit test.txt
ayano@ArnavRajput:/mnt/c/Windows/system32$ _
```

6. bg command

```
ayano@ArnavRajput:/mnt/c/windows/system32$ bg  
[1]+  sleep 500 &  
ayano@ArnavRajput:/mnt/c/windows/system32$
```

7. fg attempt / completed sleep job

```
-bash: fg: job has terminated  
[1]+  Done                  sleep 500  
ayano@ArnavRajput:/mnt/c/Windows/system32$
```

8. pstree

```
ayano@ArnavRajput:/mnt/c/Windows/system32$ pstree
systemd--agetty
      |--apt.systemd.dai--apt.systemd.dai--unattended-upgr--unattended-upgr--dpkg--uuid-runtime.po
      |                                     3*[{unattended-upgr}]
      |--apt.systemd.dai--flock
      |--cron
      |--dbus-daemon
      |--init-systemd(Ub--SessionLeader--Relay(533)--bash--pstree
      |               |--init--{init}
      |               |--login--bash
      |               |--{init-systemd(Ub}
      |--packagekitd--3*[{packagekitd}]
      |--polkitd--3*[{polkitd}]
      |--rsyslogd--3*[{rsyslogd}]
      |--systemd--(sd-pam)
      |         |--at-spi-bus-laun--dbus-daemon
      |         |               4*[{at-spi-bus-laun}]
      |         |--at-spi2-registr--3*[{at-spi2-registr}]
      |         |--dbus-daemon
      |         |--xdg-desktop-por--5*[{xdg-desktop-por}]
      |         |--xdg-desktop-por--4*[{xdg-desktop-por}]
      |         |--xdg-document-po--fusermount3
      |         |               7*[{xdg-document-po}]
      |         |--xdg-permission--3*[{xdg-permission-}]
      |--systemd-journal
      |--systemd-logind
      |--systemd-resolve
      |--systemd-timesyn--{systemd-timesyn}
      |--systemd-udev
      |--unattended-upgr--{unattended-upgr}
ayano@ArnavRajput:/mnt/c/Windows/system32$
```

9. top

```
ayano@ArnavRajput:/mnt/c/windows/system32$ top
top - 09:37:07 up 18 min, 1 user, load average: 1.03, 0.40, 0.15
Tasks: 42 total, 1 running, 41 sleeping, 0 stopped, 0 zombie
%Cpu(s): 4.4 us, 1.6 sy, 0.0 ni, 90.6 id, 3.1 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 3776.6 total, 1908.2 free, 766.6 used, 1194.6 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 3010.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
380	root	20	0	450736	213532	134056	S	33.7	5.5	1:20.37	unattended-upgr
4444	root	20	0	450736	143116	63636	S	6.7	3.7	0:00.20	unattended-upgr
3844	root	20	0	2368488	51740	25860	S	6.3	1.3	0:00.49	snapt
1	root	20	0	22004	13584	9780	S	0.3	0.4	0:07.35	systemd
159	message+	20	0	9964	5724	4700	S	0.3	0.1	0:00.79	dbus-daemon
957	root	20	0	370096	21148	18148	S	0.3	0.5	0:00.14	packagekitd
1408	root	19	-1	33932	14300	13292	S	0.3	0.4	0:00.52	systemd-journal
2	root	20	0	3180	2204	2072	S	0.0	0.1	0:00.05	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.1	0:00.00	init
158	root	20	0	4244	2744	2492	S	0.0	0.1	0:00.04	cron
166	root	20	0	17980	8964	7912	S	0.0	0.2	0:00.40	systemd-logind
182	root	20	0	3124	2036	1900	S	0.0	0.1	0:00.01	agetty
199	root	20	0	112744	25148	15200	S	0.0	0.7	0:00.21	unattended-upgr
291	root	20	0	2808	2016	1872	S	0.0	0.1	0:00.00	apt.systemd.dai
306	root	20	0	2808	2028	1872	S	0.0	0.1	0:00.00	apt.systemd.dai
531	root	20	0	3188	1108	980	S	0.0	0.0	0:00.00	SessionLeader
532	root	20	0	3204	1248	1108	S	0.0	0.0	0:00.06	Relay(533)
533	ayano	20	0	6340	5644	3648	S	0.0	0.1	0:00.24	bash
534	root	20	0	6704	4740	3956	S	0.0	0.1	0:00.01	login
580	ayano	20	0	20416	11504	9424	S	0.0	0.3	0:00.40	systemd
581	ayano	20	0	21160	3572	1848	S	0.0	0.1	0:00.00	(sd-pam)
607	ayano	20	0	6080	5328	3664	S	0.0	0.1	0:00.03	bash
715	root	20	0	2808	2024	1872	S	0.0	0.1	0:00.00	apt.systemd.dai
718	root	20	0	3124	1900	1780	S	0.0	0.0	0:00.00	flock
799	ayano	20	0	9564	5376	4756	S	0.0	0.1	0:00.32	dbus-daemon
803	ayano	20	0	382832	8044	7276	S	0.0	0.2	0:00.02	at-spi-bus-laun
810	ayano	20	0	9300	5180	4752	S	0.0	0.1	0:00.03	dbus-daemon
811	ayano	20	0	548196	16204	13512	S	0.0	0.4	0:00.06	xdg-desktop-por
816	ayano	20	0	531408	7528	6788	S	0.0	0.2	0:00.03	xdg-document-po
820	ayano	20	0	304156	6136	5636	S	0.0	0.2	0:00.00	xdg-permission-
827	root	20	0	2712	2164	2044	S	0.0	0.1	0:00.00	fusermount3
831	ayano	20	0	405996	21812	17268	S	0.0	0.6	0:00.06	xdg-desktop-por
838	ayano	20	0	236076	7988	7232	S	0.0	0.2	0:00.00	at-spi2-registr
962	polkitd	20	0	381904	8268	7320	S	0.0	0.2	0:00.13	polkitd
1469	systemd+	20	0	91036	8088	7128	S	0.0	0.2	0:00.14	systemd-timesyn
1528	root	20	0	24888	6268	5208	S	0.0	0.2	0:00.06	systemd-udev
1586	systemd+	20	0	21468	13468	11208	S	0.0	0.3	0:00.11	systemd-resolve
3259	ayano	20	0	9420	5852	3596	R	0.0	0.2	0:00.12	top

10. nice -n 10 sleep 60

```
ayano@ArnavRajput:/mnt/c/Windows/system32$ nice -n 10 sleep 60 &  
[1] 14915  
ayano@ArnavRajput:/mnt/c/Windows/system32$ ps -o pid,ni,cmd | grep "[s]leep 60"  
14915 10 sleep 60  
ayano@ArnavRajput:/mnt/c/Windows/system32$
```

4. Conclusion

The Linux process-management commands required for Lab 1 were practiced using WSL. The outputs demonstrate process creation, suspension, process listing, background execution, process termination, process-tree viewing, system monitoring with top, and priority adjustment using nice.

Note: The foreground (fg) demonstration was attempted after the sleep job had already completed, so the recorded output shows that the job had terminated rather than a successful foreground transfer.